The Unit Sphere and Degrees of Freedom:

why S^{n-1}, and what changing the norm does

Lecture 1 notes

1 What "degrees of freedom" means

Core definition: DOF = being able to *independently* move / change the value of an axis. If changing x forces y to change too, then x and y are *not* independent — they share **one** degree of freedom, not two.

The easiest model is *knobs*. Each independent knob you can turn is a degree of freedom; the number of readouts that happen to tick when you turn a knob is irrelevant.

	knobs	**independent?**	**DOF**
free $\mathbb{R}^2$	x-slider, y-slider	yes	2
circle S^1 inside $\mathbb{R}^2$	one: "how far around"	x, y move together	1

2 Why the unit sphere is S^{n-1}: each constraint strips one freedom

In $\mathbb{R}^n$ a point normally has n independent coordinates — n degrees of freedom. The norm equation

$$\|\mathbf{x}\| = 1$$

is *one* constraint, and each independent constraint kills exactly one direction of free motion, leaving $n-1$. Hence the superscript:

$$S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}, \qquad \dim = n - (\text{number of constraints}) = n - 1.$$

"S^{n-1}" says: *the surface living in* $\mathbb{R}^n$*, minus the one direction the norm equation locked down.*

	lives in	**constraint**	**dim**	**what it is**
S^0	$\mathbb{R}^1$	$\|x\| = 1$	$1 - 1 = 0$	$\{-1, +1\}$: two points, no motion
S^1	$\mathbb{R}^2$	$\|x\| = 1$	$2 - 1 = 1$	the unit circle: one tangent direction
S^2	$\mathbb{R}^3$	$\|x\| = 1$	$3 - 1 = 2$	sphere surface: lat/long

Dimension is motion between points (a continuum of neighbors), not how many points the set contains. S^0 has two elements but zero freedom, because there is no continuous road between -1 and $+1$.

3 Changing the norm changes the shape

The unit sphere is always defined the same way,

$$\{\mathbf{x} : \|\mathbf{x}\| = 1\}.$$

Keep that rule fixed and only change how $\|\cdot\|$ measures length, and the shape redraws itself — because each norm carries its own notion of what "distance 1" means. The plots below are exactly that: every point where that norm equals 1.

4 Reading the four shapes

norm	$\|\mathbf{x}\| = 1$ means	**intuition**
ℓ_2	$x_1^2 + x_2^2 = 1$	circle — as-the-crow-flies, Euclidean
ℓ_1	$\lvert x_1\rvert + \lvert x_2\rvert = 1$	diamond — taxi on a grid (Manhattan)
ℓ_∞	$\max(\lvert x_1\rvert, \lvert x_2\rvert) = 1$	square — only the largest coordinate counts
M	$x_1^2 + 2x_2^2 = 1$	ellipse — second coordinate "costs more"

Three things to hold onto:

1. **Same template, different shapes.** Always $\{\mathbf{x} : \|\mathbf{x}\| = 1\}$; only the measuring rule changes, and it reshapes the whole sphere.
2. **The shape *is* the notion of space.** In ℓ_1 the point $(1, 1)$ has norm 2; in ℓ_∞ it has norm 1. Same two coordinates, two different distances — two geometries layered on the same $\mathbb{R}^2$.
3. **Topological fact to file away:** all four are boundaries of convex, centrally symmetric sets — they're "different looks" at the same topological sphere. The *shape* (circle vs diamond vs square) is what the norm controls.

5 The made-up norm is a weighted ℓ_2 — and a seed

The ellipse is literally the ℓ_2 circle with the second coordinate counted more heavily:

$$\|\mathbf{x}\|_M = \sqrt{x_1^2 + 2x_2^2}.$$

That single "2" stretches the circle into an ellipse, and it hints that this norm comes from a hidden inner product $\langle \mathbf{x}, \mathbf{y}\rangle_M = x_1 y_1 + 2x_2 y_2$, for which $\|\mathbf{x}\|_M^2 = \langle \mathbf{x}, \mathbf{x}\rangle_M$. This is a seed — it returns as the bridge into Lecture 2's inner products.

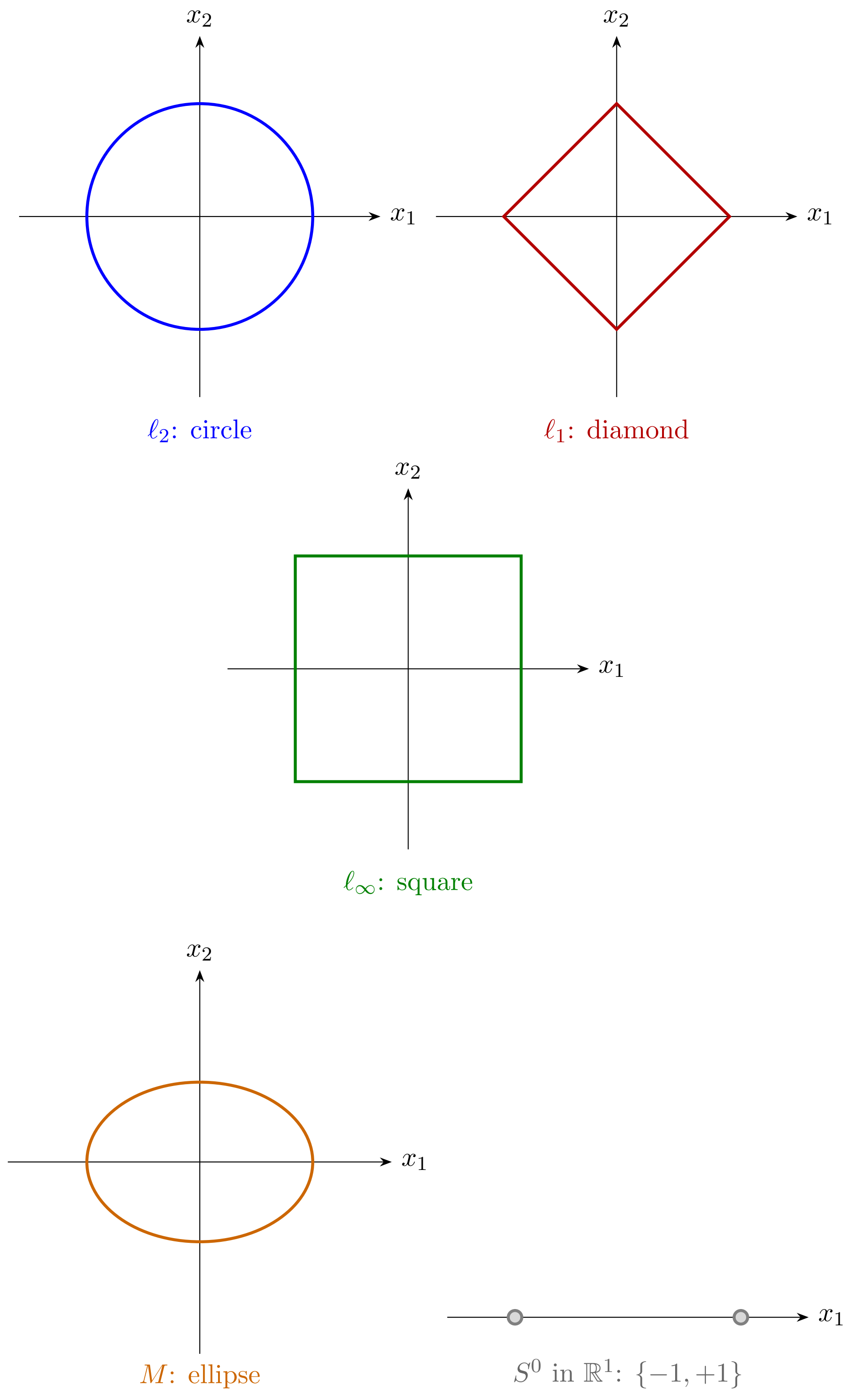

Figure 1: All plotted as $\|\mathbf{x}\| = 1$ in $\mathbb{R}^2$ (bottom-right: the zero-dimensional S^0 in $\mathbb{R}^1$ for comparison). Same equation, same coordinates, *five different geometries* of "unit circle."